

Enhancing ESL Learners' Experience and Performance through Gradual Adjustment of Video Speed during Extensive Viewing

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ABSTRACT

Learning different languages is a trend among global learners as it improves international communication. One way to learn a second language outside the classroom is through extensive viewing, which helps learners become more familiar with the target language. However, previous research has suggested that directly adjusting the video speed can disrupt learning. This study explores whether gradually adjusting the speed instead can alleviate these disruptions. We conducted an experiment (N=32) with English-as-a-second-language (ESL) participants to compare the cognitive load, flow state, and video comprehension when using the direct and gradual speed adjustment methods to accelerate or decelerate videos. Using both objective measurements, including pupil diameter, and subjective surveys, we found that the gradual method resulted in a higher flow state and improved video comprehension. These findings suggest potential directions for future research on the effects of speed adjustments on ESL learners during extensive viewing.

CCS CONCEPTS

• Human-centered computing → Empirical studies in interaction design.

KEYWORDS

Extensive Viewing; ESL Learner; Pupillometry; Playback Speed; Flow State; Cognitive Load

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1 INTRODUCTION

As technology and the internet advance, the ability to understand multiple languages becomes increasingly important for effective communication among people with different native languages. This growing internationalism of interactions has led to a greater emphasis on learning a second language (L2) or multiple languages. Especially, as English is the global lingua franca and is frequently used as a communication language on online learning platforms [44], the population of English-as-a-Second-Language (ESL) learners among L2 learners and the need to learn English as the second language has increased in recent years. Among various methods to learn a second language, *Extensive viewing* is regarded as “listening to large amounts of motivating and engaging materials which are linguistically appropriate over a while where they listen with a reasonable speed for general understanding, with a focus on meaning rather than form” [37]. Extensive viewing has many benefits, including helping learners become more fluent in speaking [1, 5, 37], increasing their vocabulary [12, 25, 26], and providing them with the opportunity to see the target language in use (e.g., recognize how facial expressions match intonation) [2, 16, 36]. Studies have also shown that ESL learners do not need to fully understand every detail of the video content, and that the general understanding and enjoyment of the material is more important for extensive viewing [15]. Additionally, extensive viewing can be highly motivating for learners as it allows them to watch engaging and authentic videos of their choice [29, 32] and the captions can also aid comprehension [18, 24, 34]. Overall, extensive viewing has recently become a valuable and engaging method for language learners [20]. However, it can be difficult for ESL learners to understand the content of videos in the target language as fluently as the native speakers [4, 49]. For example, ESL learners could face higher difficulties of understanding video content in non-native languages at an inappropriate playback speed, which can lead to problems with comprehension when learning with extensive viewing methods [30]. One strategy that ESL learners can use to optimize their learning experience in digital learning is adjusting the video speed [30, 42]. However, changing the video speed using traditional methods can have negative effects [3, 33, 51]. For example, directly adjusting the video playback speed can lead to an overwhelming cognitive load and disrupt the flow state of the learners [3].

As one of the goals for extensive viewing is to increase ESL learners' learning time outside of the classroom (e.g., watching

various videos frequently) [37, 49], it is important to consider the cognitive load of the learners and the ease of achieving a flow state when implementing extensive viewing so that they can consistently view videos on a regular basis [30, 50]. The traditional method for measuring cognitive load involves self-report questionnaires, which ask participants about their subjective experience of the task. However, these questionnaires are based on retrospective opinions and may not accurately reflect objective cognitive information [39]. Additionally, they cannot measure continuous cognitive load during the task, as measurements could only be taken before or after the task [21, 39]. Therefore, researchers have increasingly turned to physiological data, such as eye-tracking and EEG data, to study basic cognitive processes during reading and other types of information processing [35]. Physiological data can provide a more objective understanding of participants' mental and cognitive states [19, 41, 48].

In computer-assisted language learning, eye-tracking is a useful method for studying second language acquisition as it allows researchers to analyze moment-by-moment processing decisions [21, 39]. Therefore, eye-tracking metrics have been developed to access various types of perceptual and cognitive states [19, 41, 48]. As recent evidence suggests that pupil dilation can reflect one's cognitive effort and processing load in response to the stimuli, a large body of literature has proposed different indices of pupil dilation to measure human cognitive load [19, 41, 48]. Among these studies, the baseline change in pupil diameter (BCPD), the peak dilation latency (PDL), the index of pupillary activity (IPA), and the low/high index of pupillary activity (LHIPA) have been widely used to measure cognitive load [9, 10, 19, 22]. However, previous research has found mixed results in using these indices to approximate cognitive load. Some studies have supported the notion that pupil dilation can reflect cognitive load and cognitive control [8, 40, 47], while others found weak or non-significant correlations between the indices and working memory capacity and cognitive load [45, 46]. Given the mixed results, we decided to use multiple indices to examine the cognitive load to increase the reliability of our research and obtain a more comprehensive observation of the cognitive load based on pupillometry.

Motivated by the challenges ESL learners face with comprehension when using extensive viewing at unsuitable playback speeds and the drawbacks of current methods for adjusting speed, our research goal is to understand whether gradually adjusting video speed – in a smooth and not drastic way – can enable ESL learners to more easily engage in *extensive viewing* of videos made by native speakers. We expected that gradually adjusting the speed would reduce ESL learners' cognitive load, improve their comprehension, and make it easier for them to enter a flow state, compared to the direct speed adjustment used by many platforms. For this research goal, we formulated the following research questions: **RQ1. What are the effects of the gradual speed adjustment on ESL learners' cognitive load, video comprehension, and flow state compared to the direct speed adjustment? RQ2. Do the effects differ between the conditions of accelerating and decelerating videos?** To examine these research questions, we designed and conducted an empirical and quantitative experiment with 32 participants. The research findings reveal that there may be a noticeable difference in cognitive load between accelerated and

decelerated videos for ESL learners. Instead of tweaking the video speed directly, implementing a gradual pace alteration allows ESL learners to effortlessly slip into the flow state and grasp the video content with ease during extensive viewing. To put it succinctly, using a gradual method of speed adjustment proves to be a successful design, as it enhances the viewing experience for ESL learners. In the end, we conclude some research directions and improvements for our future works.

2 IMPLEMENTATION

2.1 Material Preparation

To meet the environments of extensive viewing, we collected videos about daily conversations in English (e.g., video blog) from YouTube. We randomly collected 50 English-captioned videos with the keyword “vlog” on YouTube as the initial set of materials. Then, we picked 50 clips from all the videos with a time interval between 15 to 20 seconds since the time interval for adjusting video speed in practice is often short. Furthermore, we prioritized clips in which the creator spoke while looking at the camera since auditory and visual (AV) sensory integration of speech and lip movements can improve language comprehension based on communication theory [7, 14]. After careful consideration, we eliminated 18 clips with noisy ambient sound or blurred shots or without average, continuous, and sufficient content to allow us to design three multiple-choice questions with a single correct answer. As a result, we used the remaining 32 clips as the videos for the pilot study. Next, we invited 19 volunteers to watch the videos and answer video-related questions to ensure that the difficulties of the videos were similar (Female = 6; male = 13; mean age = 20.16; averaged English level [TOEIC score] = 780.26; standard deviation [TOEIC score] = 93.01). To prevent the volunteers from not understanding the content of videos due to their English levels, we limited their English levels to at least 550 scores in the Test of English for International Communication (TOEIC), which is B1 level according to the Common European Framework of Reference for Languages (CEFR) [31]. The volunteers for the pilot study would not participate in our experiments to avoid the practice effect [6]. Finally, we selected 16 videos whose accuracy was the nearest to the mean accuracy of all videos as the materials of the experiment.

2.2 Speed Adjustment

We used editing software (VSDC) to adjust the speed of the videos. Each video was edited into four types of speed-adjusted videos based on the combinations of the direction and method of speed adjustment, which were (1) direct acceleration, (2) direct deceleration, (3) gradual acceleration, and (4) gradual deceleration. To control for the confounding variable of actual speed, we selected a speed interval between 1.0x and 1.5x that would not impact video comprehension or cognitive load. Previous research has shown that student performance is similar when video speed is between 1.0x and 1.5x, but cognitive load increases significantly when speed exceeds 1.5x [27, 28, 33, 38, 51]. Additionally, research indicates that most video viewers tend to choose the playback speed within the range of 1.0x and 1.5x [23]. Therefore, we set the speed interval from 1.0x to 1.5x to ensure that any effects on participants' performance in the study

were mainly due to the manipulation of the experimental factors, rather than increasing difficulty caused by excessive speed.

To make a video with the *direct* method of speed adjustment, we divided the video's duration time into five segments. The first two segments were in the 1.0x speed, with the other three segments in the 1.5x speed for the accelerating video group and vice versa for the decelerating group. This way, the 1.0x and 1.5x speed segments were of the same time length eventually. To make a video with the *gradual* method of speed adjustment, we divided it into three segments. The first segment aimed to render the participants familiar with the initial video speed, which was 1x or 1.5x, depending on the direction of speed adjustment. Next, since learners can adjust video speed on existing online learning platforms in minimum intervals of 0.05x, we decided to adjust the video speed by 0.05x per second to meet the realistic situation on most video-watching platforms (i.e., accelerating or decelerating video speed by 0.05x per second). Hence, the video speed in the second segment accelerated from 1x to 1.5x or decelerated from 1.5x to 1x gradually (e.g., 1x, 1.05x, to 1.45x, 1.5x times). Finally, the video speed remained at 1x or 1.5x speed in the third segment based on the direction of speed adjustment. We edited the 16 unadjusted videos into the 64 adjusted videos (16 unadjusted videos $\times$ two directions of speed adjustment $\times$ two methods of speed adjustment). Each participant watched 16 adjusted videos with the same direction of speed adjustment, with eight videos in the gradual method and the other eight videos in the direct method. These videos were displayed in random order. Our main goal was to compare the effects of gradual and direct speed adjustment on ESL learners rather than designing a new method for adjusting video playback speed. We currently offer a practical solution for adjusting video playback speed. However, the possibility of gradually adjusting speed throughout a video is an area that could be explored in future research.

3 EXPERIMENT DESIGN

3.1 Environment Setup and Participants

We used the Eye Tribe ET1000 at 30 Hz to record participants' eye movements and pupil diameter. We placed the eye tracker 450 mm in front of the participants, between them and the screen. Each participant was seated comfortably in a sufficiently illuminated (140 lux) room on a stable chair at a distance of 600 mm from the stimulus screen (HP M24f FHD Monitor, 23.8 in, 1980x1080 pixels, 60 Hz refresh rate). We stabilized the participants' heads with a chin rest and asked them to wear headphones (see Figure 1(a)) to ensure they heard the video's sound clearly with similar loudness.

We recruited 32 non-native English speakers as participants. All the participants are above 20 years old with normal vision. There are 13 men and 19 women, with an average age of 22.19 years old ($SD = 2.83$) and an average English level on the TOEIC of 781.41 ($SD = 91.64$). Similar to the requirements for the volunteers in the material preparation, all the participants must achieve at least 550 scores TOEIC, which is B1 level according to the Common European Framework of Reference for Languages (CEFR) [31].

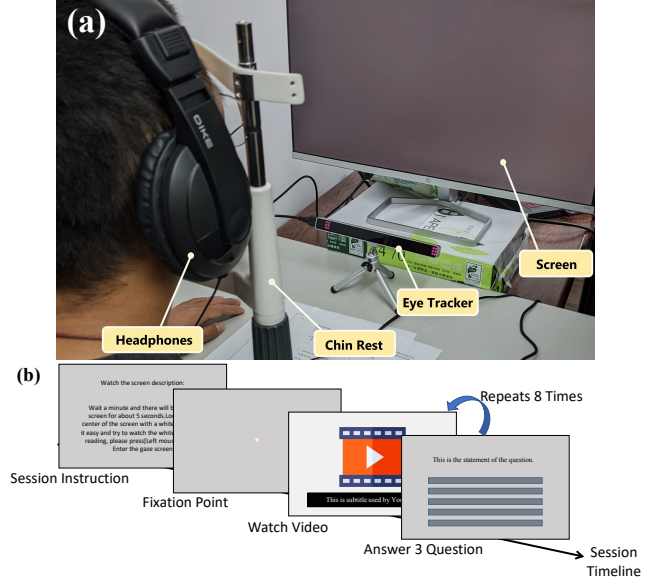


Figure 1: (a) Participants wore headphones, rested their heads on a chin rest, and watched stimuli on a screen while an eye tracker recorded their pupil diameter. (b) Participants completed two sessions, each involving reading instructions, watching a fixation point for 5 seconds, and then answering 3 video-related questions 8 times.

3.2 Condition and Procedure

Table 1: Experiment conditions. The method of speed adjustment is the within-subject factor, and the direction is the between-subject factor. Each participant conducted different methods of speed adjustment in random order.

Condition	Phase one	Phase two
C1	Direct acceleration	Gradual acceleration
C2	Gradual acceleration	Direct acceleration
C3	Direct deceleration	Gradual deceleration
C4	Gradual deceleration	Direct deceleration

Our experiment employed a 2×2 mixed factorial design, with the *direction* of speed adjustment (accelerating or decelerating) as a between-subjects factor and the *method* of speed adjustment (direct or gradual) as a within-subjects factor. We assigned the participants evenly into two groups of the direction of speed adjustment by matching their English CEFR levels translated from their TOEIC score to ensure that the English abilities in each group were similar. In the experiment, each group of participants completed the two levels in the within-subject factor (the method of speed adjustment) randomly, following the experimental design shown in Table 1, to minimize any order effects.

Our experiment comprised three sessions: the practice session, the first session, and the second session. The first and second sessions are for each phase in the experiment condition. After we

obtained a good level of eye-tracker calibration, we asked participants to watch the instruction about the upcoming session. Next, the participants were asked to stare at the yellow dot at the center of the screen for five seconds to collect the pupil size data that would be used as the baseline in measuring cognitive load. Then, three non-adjusted videos were displayed along with three multiple-choice questions in English during the practice session to ensure the participants to have understood the experiment process. After finishing the practice session, the participants continued to the following two sessions according to each phase in the experiment condition. The procedure for the practice session and the following two sessions were similar, and the differences were the number of videos displayed and the speed adjustment of videos (as shown in Figure 1(b)). The order in which the 16 videos were displayed was randomized for each participant and scattered randomly in the two sessions. The speed adjustment of videos in each session was based on the manipulation of each phase in the experiment condition shown in Table 1. After the participants finished the first or second session, they were asked to complete the self-reported survey of NASA-TLX [17] and flow state scale (FSS) [11], which we explain in the next section. They were free to rest between the two sessions for a maximum of five minutes. Our institution's IRB approved the experiment procedure and design. The average length of each experiment was around 45 minutes. Each participant was compensated 200 NTD (equivalent to 6.48 USD) for their time after the experiment.

3.3 Measurement

Pupillometry Indices of Cognitive Load. We used four indices (BCPD, PDL, IPA, and LHIPA) to measure the participants' pupil dilation data when they watched the eight videos in each phase for the continuous cognitive load during watching. Thus, for each participant, we computed 64 pupil dilation data points (the four indices $\times$ the eight videos $\times$ the two phases). To compute the BCPD, we used the method detailed in [22]. We calculated the average pupil diameter in the fixation screen of the test session as the baseline for the BCPD. Then, we computed the average difference between the pupil diameter and the average over the baseline for each pupil dilation data point recorded in each video to get the BCPD. To compute PDL, we used the procedure described in [19], calculating the time interval from the start of the videos to the time when pupil diameter reaches the highest peak. We used the method for computing IPA from [10] and calculated the LHIPA using the method from [9]. We adjusted the wavelet transformation used in the IPA and LHIPA to the Daubechies-4 wavelet, as recommended in [10], because the eye tracker in our experiment has a relatively low sampling rate. In general, a higher BCPD, PDL, and IPA value reflect a higher cognitive load, while a higher LHIPA value reflects a lower cognitive load.

Self-report Measures. In addition to measuring the cognitive load using pupillometry indices, we also collected participants' self-reported cognitive load by asking them to complete the NASA-TLX scale [17] after each phase. After the participants finished the NASA-TLX form, they completed the flow short scale (FSS) [11]. The FSS is constructed by three aspects: participants' fluency, absorption, and perceived importance. We dropped the perceived importance aspect

as we deemed it irrelevant to the experiment. The FSS is a 7-point Likert scale and the questionnaire ranges from one ("Not at all") to seven ("Very much"). We took the mean score of the six questions (Q2, Q4, Q5, Q7, Q8, Q9) as the participants' fluency with the videos in each session. To determine their absorption of the videos, we calculated the mean score of the remaining four questions (Q1, Q3, Q6, Q10).

Behavioral Measures. We measured the participants' video comprehension by checking the accuracy of their answers to the video-related questions. We measured each participant's accuracy by dividing the number of correct answers by 24 questions (eight videos $\times$ three questions per video) for each phase. Accordingly, each participant had two sets of accuracy for two phases in the experiment. Next, we recorded the participants' answering times separately to measure how much time they spent answering the three questions after watching each video. Hence, there were 16 answering times (eight videos $\times$ two phases) for each participant.

4 RESULT AND DISCUSSION

4.1 Result

Cognitive Load. Two-way mixed ANOVA was used to analyze the NASA-TLX, and three-way mixed ANOVA was used to analyze the other pupillometry indices, as the order of the video also had an impact on these pupillometry indices. There were no significant differences in any indices between the gradual method and the direct one. Neither was the interaction between the two methods and the two directions regardless of how cognitive load indices were measured. Furthermore, the two directions (e.g., the acceleration and deceleration) had a significant difference in BCPD ($F(1,5) = 6.883, p < 0.05$) but did not have significant differences on any other indices. It appears that decreasing the speed might have a tendency to reduce cognitive load more than increasing the speed.

Flow State. The two dimensions of flow, fluency and absorption, were analyzed using different statistical methods. Two-way mixed ANOVA was used to analyze absorption, and two-way ANCOVA was used to analyze fluency. This was because we found that English proficiency level had an impact on fluency. There are significantly higher scores when the gradual method is used than when the direct one is used on both fluency ($F(1,6) = 9.207, p < 0.05$) and absorption ($F(1,30) = 5.384, p < 0.05$). It shows that using the gradual method is easier to get into the flow state. Surprisingly, there is no significance on absorption between acceleration and deceleration, but deceleration has significantly higher scores on fluency than acceleration ($F(1,6) = 8.301, p < 0.05$). This result demonstrates that decelerating video had different effects on different dimensions of the flow state in our experiments. Although participants perceived higher fluency in the conditions of decelerating videos, their perceived absorption remained at a similar level. Last, there are no significant interactions between the two factors.

Behavioral Results. While accuracy was analyzed using two-way mixed ANOVA, answering time was analyzed using three-way mixed ANOVA. This is because answering time was calculated for each video separately. Our results indicate that the gradual method for adjusting playback speed leads to significantly higher accuracy compared to the direct method ($F(1,30) = 6.634, p < 0.05$). However, there was no significant difference in answering time between the

Table 2: Descriptive statistics of cognitive load were measured by self-report survey and pupillometry indices, with "Acc." representing acceleration conditions and "Dec." representing deceleration conditions.

Mean (SE)		NASA	BCPD	PDL	IPA	LHIPA
Acc.	Direct	60.98 (4.53)	3.06 (.256)	8822 (469)	.121 (.006)	1.29 (.042)
	Gradual	61.25 (4.71)	3.08 (.311)	9246 (457)	.122 (.006)	1.35 (.053)
Dec.	Direct	61.60 (4.13)	2.97 (.265)	8336 (390)	.133 (.006)	1.36 (.044)
	Gradual	60.98 (4.43)	3.27 (.230)	7990 (403)	.125 (.006)	1.35 (.045)

two methods. This suggests that gradual changes in speed may help learners better comprehend the video content. Moreover, deceleration was found to result in shorter answering times compared to acceleration ($F(1,27) = 5.896, p < 0.05$), but there was no significant difference in accuracy between the two directions. This finding indicates that decelerating the video speed may help learners absorb information more quickly. No significant interactions were found between the two factors. Overall, our findings suggest that gradual changes in speed and deceleration may be more effective for learners in improving their video comprehension and answering speed.

4.2 Effects of Speed-Adjustment Method and Direction

Our study found that ESL participants who used the gradual method had higher accuracy scores than those who used the direct method, suggesting that the gradual method may enhance video comprehension during extensive learning. However, we did not observe significant effects of speed-adjustment methods on cognitive load measured by pupillometry and self-report surveys. The lack of significance in cognitive load may be attributed to various factors (e.g., the complex interaction between participants' germane and extraneous load [43], which are categories of cognitive load). Further research is needed to identify the accurate reasons. In addition, as shown in the results of the flow state, the gradual method made the participants easier to get into the flow state than the direct one, which represents that the gradual change of adjusting speed could reduce the interference caused by changing speed. To sum up, the method of gradual speed adjustment could significantly improve ESL learners' video comprehension and flow state in the scenario of extensive viewing.

Next, our results revealed that the directions of speed adjustment (i.e., acceleration and deceleration) have different effects on ESL learners. BCPD results indicated lower cognitive load in deceleration compared to acceleration conditions, consistent with prior research [27]. However, other indices did not show significant differences between speed adjustment directions. Further investigation is needed. Nevertheless, other measures supported decelerated videos' benefits for ESL learners. The participants have higher fluency and shorter answering times in the deceleration conditions compared to the acceleration ones. This finding could indicate that the deceleration could ease the difficulties learners experienced during watching English videos and possibly quicken the process of knowledge retrieved when answering the video-related questions. Lastly, since we did not observe the interaction between the method and direction of the speed adjustment, the current preliminary result might indicate that the benefits of the gradual method would not

be limited to the specific direction of speed adjustment. However, more studies are needed to verify the above conclusion.

5 CONCLUSION AND FUTURE WORK

Given that extensive viewing is an effective way for ESL learners to learn English, our study aimed to investigate the impact of the different methods of adjusting video speed on their cognitive load, flow state, and video comprehension. Our findings revealed that using a gradual method for adjusting speed, rather than a direct one, did improve the ESL learners' experience and performance when they were engaged in extensive learning. This improvement was observed in terms of self-report flow state and video comprehension. Additionally, our findings suggest that watching decelerated videos may be less challenging than watching accelerated videos, as indicated by a smoother flow state, faster answering time, and possibly lower cognitive load. Based on the current results, we conclude that the use of the gradual method is an effective alternative, and it can reduce the interference caused by the method of directly adjusting video speed.

As our current study was conducted in the context of automatically adjusting playback speed, it might be a lack of ecological validity since learners usually adjust the playback speed of video manually and freely. To increase the ecological validity, in future studies, we plan to investigate if the same effect can be observed when the method is applied in a context where ESL learners manually adjust the video speed during extensive viewing. Additionally, we are interested in exploring if the effect holds true across a range of different speed adjustments. Moreover, we want to build a video-watching prototype platform with the gradual speed-adjustment method and conduct field studies with the platform to observe the effect of the gradual method on ESL learners in a more realistic setting. Lastly, the current gradual method employed linear adjustments. However, the research in psychophysics has indicated that the relationship between changes in external stimuli and changes in psychological experience is not always proportional [13]. Therefore, we also want to design and evaluate a gradual method that utilizes curved adjustments to make the differences in speed less noticeable. By considering more important design elements of video speed adjustment in the context of extensive viewing, we hope to offer a comprehensive design space for how various speed-adjustment approaches can effectively benefit ESL learners in the future.

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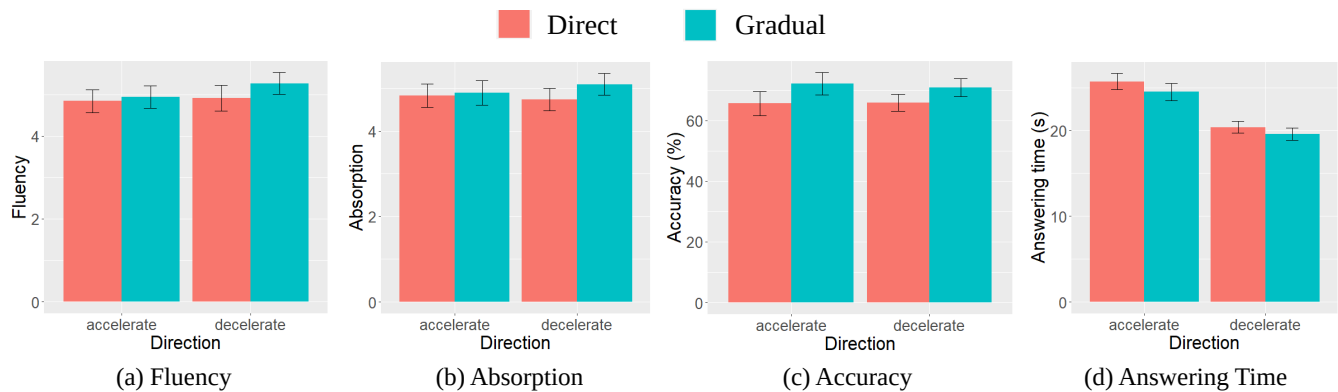


Figure 2: (a) and (b) show self-reported fluency and absorption scores on FSS, respectively. (c) displays accuracy percentage, and (d) shows answering time in seconds. The error bars represent the standard error.

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REFERENCES

- [1] Lumturie Bajrami and Merita Ismaili. 2016. The role of video materials in EFL classrooms. *Procedia-Social and Behavioral Sciences* 232 (2016), 502–506. <https://doi.org/10.1016/j.sbspro.2016.10.068>
- [2] Betül Bal-Gezegin. 2014. An investigation of using video vs. audio for teaching vocabulary. *Procedia-Social and Behavioral Sciences* 143 (2014), 450–457. <http://hdl.handle.net/10063/2870>
- [3] Stefan Berti and Erich Schröger. 2006. Visual distraction: a behavioral and event-related brain potential study in humans. *NeuroReport* 17, 2 (2006), 151–155. <https://doi.org/10.1097/01.wnr.0000195669.07467.e1>
- [4] Eileen K Blau. 1990. The effect of syntax, speed, and pauses on listening comprehension. *TESOL quarterly* 24, 4 (1990), 746–753. <https://doi.org/10.2307/3587129>
- [5] Anna C-S Chang. 2016. Teaching L2 listening: In and outside the classroom. In *English language teaching today*. Springer, Berlins, 111–125. https://doi.org/10.1007/978-3-319-38834-2_9
- [6] Larry B Christensen, Burke Johnson, Lisa Anne Turner, and Larry B Christensen. 2011. *Research methods, design, and analysis*. Pearson Boston, Boston. <https://doi.org/10.5772/intechopen.85731>
- [7] Herbert H Clark. 1996. *Using language*. Cambridge university press, Cambridge.
- [8] Russell A Cohen Hoffing, Nina Lauharatanahirun, Daniel E Forster, Javier O Garcia, Jean M Vettel, and Steven M Thurman. 2020. Dissociable mappings of tonic and phasic pupillary features onto cognitive processes involved in mental arithmetic. *PLoS one* 15, 3 (2020), e0230517. <https://doi.org/10.1371/journal.pone.0230517>
- [9] Andrew T Duchowski, Krzysztof Krejtz, Nina A Gehrer, Tanya Bafna, and Per Bækgaard. 2020. The low/high index of pupillary activity. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3313831.3376394>
- [10] Andrew T Duchowski, Krzysztof Krejtz, Izabela Krejtz, Cezary Biele, Anna Niedzielska, Peter Kiefer, Martin Raubal, and Ioannis Giannopoulos. 2018. The index of pupillary activity: Measuring cognitive load vis-à-vis task difficulty with pupil oscillation. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3173856>
- [11] Stefan Ed Engesser. 2012. *Advances in flow research*. Springer Science+ Business Media, New York, NY, US. <https://doi.org/10.1007/978-1-4614-2359-1>
- [12] Yanxue Feng and Stuart Webb. 2020. Learning vocabulary through reading, listening, and viewing: Which mode of input is most effective? *Studies in Second Language Acquisition* 42, 3 (2020), 499–523. <https://doi.org/10.1017/S0272263119000494>
- [13] George A Gescheider. 2013. *Psychophysics: the fundamentals*. Psychology Press, London. <https://doi.org/10.4324/9780203774458>
- [14] Susan Goldin-Meadow. 2005. *Hearing gesture: How our hands help us think*. Harvard University Press, Cambridge, USA.
- [15] Suzanne Graham and Denise Santos. 2015. *Strategies for second language listening: Current scenarios and improved pedagogy*. Springer, Berlin. <https://books.google.com.tw/books?id=rlveCgAAQBAJ>
- [16] Jeremy Harmer. 2001. *The practice of English language teaching*. Pearson, London/New York.
- [17] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, Amsterdam, 139–183. [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)
- [18] Abdolmajid Hayati and Firooz Mohmedi. 2011. The effect of films with and without subtitles on listening comprehension of EFL learners. *British Journal of Educational Technology* 42, 1 (2011), 181–192. <https://doi.org/10.1111/j.1467-8535.2009.01004.x>
- [19] Jamie N Hershaw and Mark L Ettenhofer. 2018. Insights into cognitive pupillometry: Evaluation of the utility of pupillary metrics for assessing cognitive load in normative and clinical samples. *International Journal of Psychophysiology* 134 (2018), 62–78. <https://doi.org/10.1016/j.ijpsycho.2018.10.008>
- [20] Francisca M Ivone and Willy A Renandya. 2019. Extensive listening and viewing in ELT. *Teflin Journal* 30, 2 (2019), 237–256. <https://doi.org/10.15639/teflinjournal.v30i2/237-256>
- [21] Marcel A Just and Patricia A Carpenter. 1980. A theory of reading: from eye fixations to comprehension. *Psychological review* 87, 4 (1980), 329. <https://doi.org/10.1037/0033-295X.87.4.329>
- [22] Krzysztof Krejtz, Andrew T Duchowski, Anna Niedzielska, Cezary Biele, and Izabela Krejtz. 2018. Eye tracking cognitive load using pupil diameter and micro-saccades with fixed gaze. *PLoS one* 13, 9 (2018), e0203629. <https://doi.org/10.1371/journal.pone.0203629>
- [23] David Lang, Guanling Chen, Kathy Mirzaei, and Andreas Paepcke. 2020. Is faster better? A study of video playback speed. In *Proceedings of the tenth international conference on learning analytics & knowledge*. ACM, New York, 260–269.
- [24] Mina Lee, Beverly Roskos, and David R Ewoldsen. 2013. The impact of subtitles on comprehension of narrative film. *Media Psychology* 16, 4 (2013), 412–440. <https://doi.org/10.1080/15213269.2013.826119>
- [25] Wang Li and Willy A Renandya. 2012. Effective approaches to teaching listening: Chinese EFL teachers' perspectives. *The Journal of Asia TEFL* 9, 4 (2012), 79–111.
- [26] Ahmed Masrai. 2020. Can L2 phonological vocabulary knowledge and listening comprehension be developed through extensive movie viewing? The case of Arab EFL learners. *International Journal of Listening* 34, 1 (2020), 54–69. <https://doi.org/10.1080/10904018.2019.1582346>
- [27] Chuan-Yu Mo, Chengliang Wang, Jian Dai, and Peiqi Jin. 2022. Video playback speed influence on learning effect from the perspective of personalized adaptive learning: A study based on cognitive load theory. *Frontiers in Psychology* 13 (2022), 1–11.
- [28] Dillon H Murphy, Kara M Hoover, Karina Agadzanyan, Jesse C Kuehn, and Alan D Castel. 2022. Learning in double time: The effect of lecture video speed on immediate and delayed comprehension. *Applied Cognitive Psychology* 36, 1 (2022), 69–82. <https://doi.org/10.1002/acp.3899>
- [29] Asnawi Muslem, Faisal Mustafa, Bustami Usman, and Aulia Rahman. 2017. The Application of Video Clips with Small Group and Individual Activities to Improve Young Learners' Speaking Performance. *Teaching English with Technology* 17, 4 (2017), 25–37.
- [30] Naoto Nishida, Hinako Nozaki, and Buntarou Shizuki. 2022. Laugh at Your Own Pace: Basic Performance Evaluation of Language Learning Assistance by Adjustment of Video Playback Speeds Based on Laughter Detection. In *Proceedings of the Ninth ACM Conference on Learning@ Scale*. ACM, New York, 368–373. <https://doi.org/10.1145/3491140.3528299>

- [31] Council of Europe. Council for Cultural Co-operation. Education Committee. Modern Languages Division. 2001. *Common European framework of reference for languages: Learning, teaching, assessment*. Cambridge University Press, Cambridge. <https://books.google.com.tw/books?id=PygQ8Gk4k4YC>
- [32] Mehmet Özgen and Nazlı Gündüz. 2020. Authentic Captioned Sitcom as Listening Comprehension Material in English Language Teaching. *ELT Research Journal* 9, 2 (2020), 167–193.
- [33] Ray Pastore. 2012. The effects of time-compressed instruction and redundancy on learning and learners' perceptions of cognitive load. *Computers & Education* 58, 1 (2012), 641–651. <https://doi.org/10.1016/j.compedu.2011.09.018>
- [34] Elke Peters, Eva Heynen, and Eva Puimège. 2016. Learning vocabulary through audiovisual input: The differential effect of L1 subtitles and captions. *System* 63 (2016), 134–148. <https://doi.org/10.1016/j.system.2016.10.002>
- [35] Keith Rayner. 2009. The 35th Sir Frederick Bartlett Lecture: Eye movements and attention in reading, scene perception, and visual search. *Quarterly journal of experimental psychology* 62, 8 (2009), 1457–1506. <https://doi.org/10.1080/17470210902816461>
- [36] Willy A Renandya and Thomas SC Farrell. 2011. 'Teacher, the tape is too fast!' Extensive listening in ELT. *ELT journal* 65, 1 (2011), 52–59. <https://doi.org/10.1093/elt/ccq015>
- [37] Willy A Renandya and George M Jacobs. 2016. Extensive reading and listening in the L2 classroom. In *English language teaching today*. Springer, Berlin, 97–110. https://doi.org/10.1007/978-3-319-38834-2_8
- [38] Albert D Ritzhaupt, Raymond Pastore, and Robert Davis. 2015. Effects of captions and time-compressed video on learner performance and satisfaction. *Computers in Human Behavior* 45 (2015), 222–227. <https://doi.org/10.1016/j.chb.2014.12.020>
- [39] Leah Roberts and Anna Siyanova-Chanturia. 2013. Using eye-tracking to investigate topics in L2 acquisition and L2 processing. *Studies in Second Language Acquisition* 35, 2 (2013), 213–235. <https://doi.org/10.1017/S0272263112000861>
- [40] Adi Shechter and David L Share. 2021. Keeping an eye on effort: A pupillometric investigation of effort and effortlessness in visual word recognition. *Psychological Science* 32, 1 (2021), 80–95. <https://doi.org/10.1177/0956797620958638>
- [41] Sylvain Sirois and Julie Brisson. 2014. Pupillometry. *Wiley Interdisciplinary Reviews: Cognitive Science* 5, 6 (2014), 679–692. <https://doi.org/10.1002/wcs.1323>
- [42] Kousuke SUGAI, Shigeru Yamane, and Kazuo Kanzaki. 2016. The time domain factors affecting EFL learners' listening comprehension: A study on Japanese EFL learners. *ARELE: Annual Review of English Language Education in Japan* 27 (2016), 97–108. https://doi.org/10.20581/arele.27.0_97
- [43] John Sweller. 2010. Element interactivity and intrinsic, extraneous, and germane cognitive load. *Educational psychology review* 22, 2 (2010), 123–138. <https://doi.org/10.1007/s10648-010-9128-5>
- [44] Peter Trudgill and Jean Hannah. 2013. *International English: A guide to the varieties of standard English*. Routledge, Oxfordshire. <https://doi.org/10.4324/9780203785225>
- [45] Nash Unsworth, Ashley L Miller, and Matthew K Robison. 2021. Is working memory capacity related to baseline pupil diameter? *Psychonomic Bulletin & Review* 28, 1 (2021), 228–237. <https://doi.org/10.3758/s13423-020-01817-5>
- [46] Nash Unsworth, Matthew K Robison, and Ashley L Miller. 2019. Individual differences in baseline oculometrics: Examining variation in baseline pupil diameter, spontaneous eye blink rate, and fixation stability. *Cognitive, Affective, & Behavioral Neuroscience* 19, 4 (2019), 1074–1093. <https://doi.org/10.3758/s13415-019-00709-z>
- [47] Pauline van der Wel and Henk van Steenbergen. 2018. Pupil dilation as an index of effort in cognitive control tasks: A review. *Psychonomic bulletin & review* 25, 6 (2018), 2005–2015. <https://doi.org/10.3758/s13423-018-1432-y>
- [48] Pascal WM Van Gerven, Fred Paas, Jeroen JG Van Merriënboer, and Henk G Schmidt. 2004. Memory load and the cognitive pupillary response in aging. *Psychophysiology* 41, 2 (2004), 167–174. <https://doi.org/10.1111/j.1469-8986.2003.00148.x>
- [49] Stuart Webb. 2015. Extensive viewing: Language learning through watching television. In *Language learning beyond the classroom*. Routledge, Oxfordshire, 159–168.
- [50] C Whitson and J Consoli. 2009. Flow theory and student engagement. *Journal of Cross-Disciplinary Perspectives in Education* 2, 1 (2009), 40–49.
- [51] Xiaozhe Yang, Lin Lin, Yi Wen, Pei-Yu Cheng, Xue Yang, and Yunjo An. 2020. Time-compressed audio on attention, meditation, cognitive load, and learning. *Educational Technology & Society* 23, 3 (2020), 16–26. <https://www.jstor.org/stable/26926423>